

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Project Name: Plugging into the Future: An Exploration of Electricity Consumption Patterns

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Access	Open Tableau dashboard through Flask web application
FR-2	State-wise Analysis	View electricity consumption by state
FR-3	Region-wise Analysis	Filter and compare North, South, East, West, Central, North-East regions
FR-4	Year-wise Comparison	Compare electricity consumption for 2019 and 2020
FR-5	Interactive Visualizations	Display India map, bar charts, and pie charts
FR-6	Data Filtering	Filter data by state, region, and year
FR-7	Insight Generation	Identify highest and lowest electricity-consuming states
FR-8	Report Visualization	Present consumption trends through dashboards

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, interactive, and user-friendly dashboard interface.
NFR-2	Security	Secure access to the Flask application and dashboard.
NFR-3	Reliability	Consistent and accurate visualization of electricity data.

NFR-4	Performance	Dashboard should load and respond quickly.
NFR-5	Availability	Dashboard should be accessible whenever required.
NFR-6	Scalability	Support additional years, states, and datasets in the future.